# Mathematics

International Physics Olympiad · Worked Solutions

Introductory study guide · 10 original problems

> Organized around the IPhO syllabus. This guide is not exhaustive; consult the syllabus study PDF and the IPhO syllabus webpage for the full scope and qualifications.

These worked solutions accompany the original MAIS PAC problems. Check your assumptions and units.

1. Solve $x^2 - 5x + 6 = 0$.

   **Solution.** Factor as $(x-2)(x-3) = 0$; $x = 2$ or $3$. A physical problem may rule out a root.

2. Solve e^(−t/2) = 1/4 for t.

   **Solution.** Taking logarithms gives $-t/2 = -\ln 4$, hence $t = 2 \ln 4$.

3. Find the area of a circle of radius 3 m.

   **Solution.** $A = \pi r^2 = 9\pi$ m².

4. For a = (1, 2) and b = (3, 4), find a · b.

   **Solution.** $a \cdot b = 1\times3 + 2\times4 = 11$.

5. Express $(1+i)^2$ in algebraic form.

   **Solution.** $(1+i)^2 = 1 + 2i + i^2 = 2i$.

6. Measurements are 2, 4 and 6. Find their mean and sample standard deviation.

   **Solution.** The mean is 4. The squared deviations sum to 8, so $s = \sqrt{(8/2)} = 2$.

7. If $x(t) = 3t^2 + 2$, find velocity and displacement from t = 0 to t = 2.

   **Solution.** $v = dx/dt = 6t$. The displacement is $x(2) - x(0) = 12$.

8. Approximate $\sqrt{1.04}$ to first order in 0.04.

   **Solution.** Using $(1+x)^{1/2} \approx 1+x/2$ gives 1.02; this is close to 1.0198.

9. Challenge 1 — A positive solution of $x + \varepsilon x^2 = 1$ is close to 1 when $|\varepsilon|$ is small. Find it to first order in $\varepsilon$.

   **Solution.** Put $x = 1 + a\varepsilon + O(\varepsilon^2)$. Substitution gives $1 + (a+1)\varepsilon + O(\varepsilon^2) = 1$, so $a = -1$ and $x = 1-\varepsilon + O(\varepsilon^2)$. This is the branch continuous at $\varepsilon = 0$.

10. Challenge 2 — An ideal series RC circuit has resistance R, capacitance C and angular frequency $\omega$. Find the magnitude of the voltage across the capacitor divided by the source voltage.

    **Solution.** With the e^(iωt) convention, ZC = 1/(iωC). The voltage-divider ratio is ZC/(R+ZC) = 1/(1+iωRC). Its magnitude is $1/\sqrt{(1+\omega^2R^2C^2)}$. It approaches 1 at low frequency and 0 at high frequency.